MASTER FUNCTIONAL SKILLS

OFFICIAL EXAM SIMULATION SERIES

OFFICIAL SIMULATION PACK • 8 COMPLETE TESTS

MOCK PACK

MOCK EXAMS

8 FULL EXAM PAPERS

MATHS LEVEL 2

Authentic Section A (Non-Calc) & Section B (Calc) booklets with complete mark schemes

8 Full Papers | Section A & B | Mark Schemes | Method Criteria

PEARSON EDEXCEL & CITY & GUILDS ALIGNED • 2026/2027

OFFICIAL STUDY MANUAL

MATHS LEVEL 2

Functional Skills Mathematics • Level 2

Official 6-Mock Examination Pack • Complete Papers, Step-by-Step Mark Schemes & Process Mark Rubrics

This book belongs to:	Exam date:	Target level:
_____	_____	_____

What is inside

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How to use this book

Read
Read the rule. Study the picture.

Try
Write your answers on the lines.

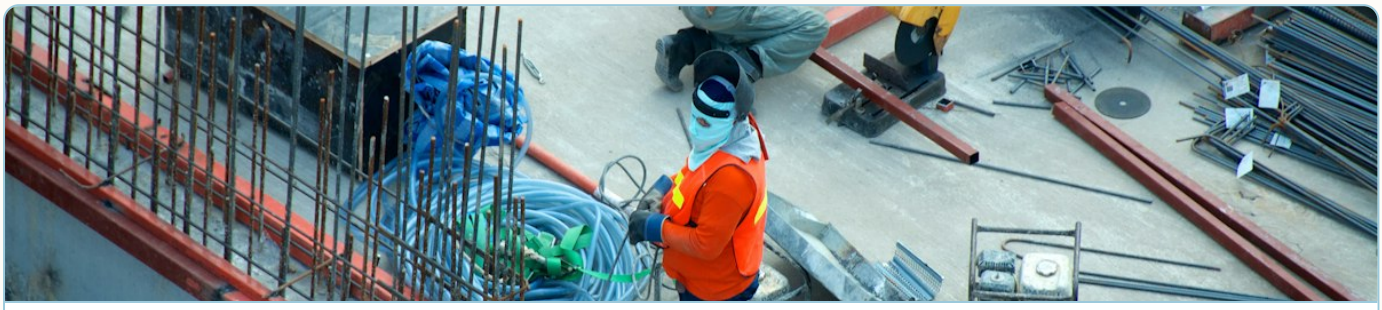
Check
Mark it with the answers at the back.

Remember
Learn the Golden Rule by heart.

F1

FORMULA REFERENCE SHEET • PART 1

Level 2 Essential Formulae: Geometry & Shapes



Geometry calculations underpin construction and engineering exams.

▶ Geometric Formulae (Memorise for Section A & B)

Circumference of Circle:	$C = \pi d = 2\pi r$ (use $\pi = 3.14$ or 3.142)
Area of Circle:	$A = \pi r^2$ (always square radius, NEVER multiply by 2!)
Area of Cylinder (Curved):	Curved Surface Area = $\pi dh = 2\pi rh$
Volume of Cylinder:	$V = \pi r^2 h$ (Area of cross-section × height)
Volume of Triangular Prism:	V = (½ × base × height) × length
Pythagoras Theorem:	$a^2 + b^2 = c^2$ (where c is the hypotenuse)

✓ Golden Geometry Rule

For semi-circles, divide circumference or area by 2. Add diameter if calculating perimeter.

! Examiner Trap!

Remember: πr^2 is NOT $(\pi r)^2$. Square radius first, then multiply by π.

✎ Personal Geometry Notes & Formula Memory Triggers

F2

FORMULA REFERENCE SHEET • PART 2

Level 2 Essential Formulae: Measures & Statistics

Measures, Rates & Statistics Matrix

Speed, Distance & Time:	`Speed = Distance ÷ Time | Time = Dist ÷ Speed`
Density, Mass & Volume:	`Density = Mass ÷ Volume | Mass = Density × Volume`
Percentage Change:	`% Change = (Actual Change ÷ Original Value) × 100`
Reverse Percentage:	`Original Value = New Value ÷ Decimal Multiplier`
Estimated Mean (Grouped):	`Estimated Mean = Σ(midpoint × frequency) ÷ Σ(frequency)`
Combined Probability:	`P(A and B) = P(A) × P(B)` (for independent events)



Abdul

Time Trap Alert! 2 hours 15 minutes is 2.25 hours, NOT 2.15 hours! Always convert minutes to decimal hours by dividing by 60 before using speed formulas.

✎ Key Conversion Facts (e.g. 1 m³ = 1,000 litres, 1 gallon = 4.55 litres)

EXAMINER STRATEGY GUIDE

Exam Technique & Securing Process Marks



Every intermediate step secures process marks even if your final answer is wrong.

Pearson Mark Classification System

[P] Process Marks: Awarded for choosing the correct strategy (e.g. recognizing a 20% discount means multiplying by 0.80).

[A] Accuracy Marks: Awarded strictly for the correct numerical calculation.

[M] Method Marks: Awarded for writing out standard formulas with given values substituted correctly.

[B] Independent Marks: Awarded for correct units (£, m², kg), rounding, or valid written checks.

How to Write a Guaranteed 1-Mark 'Check'

When asked to *'Show a check of your answer'*, use one of two methods:

1. Reverse Operation (Inverse): If `45 × 12 = 540`, write `Check: 540 ÷ 12 = 45`.

2. Estimation Check: Round inputs to 1 significant figure: `Check: 48.2 × 9.8 ≈ 50 × 10 = 500`.

Trap: Simply rewriting `12 × 45 = 540` scores **0 marks**!

✎ My Personal Exam Rules & Timing Checklist

SCORES & PROGRESS MONITORING

TR Diagnostic Mock Performance Tracker

Pearson Edexcel Level 2 Pass Benchmark: 38–42 Marks (approx. 60–65%)

Paper	Sec A (/16)	Sec B (/48)	Total (/64)	% Score	Status	Date
Mock Paper 1	/16	/48	/64	%	[ ] Pass	
Mock Paper 2	/16	/48	/64	%	[ ] Pass	
Mock Paper 3	/16	/48	/64	%	[ ] Pass	
Mock Paper 4	/16	/48	/64	%	[ ] Pass	
Mock Paper 5	/16	/48	/64	%	[ ] Pass	
Mock Paper 6	/16	/48	/64	%	[ ] Pass	

✎ Target Topics for Post-Mock Revision & 1-to-1 Tutoring

P1.A

PAPER 1: FACILITIES MANAGEMENT & EVENT OPERATIONS

Mock Paper 1 • Section A (Non-Calculator)

Section A Candidate Instructions

Duration: 25 minutes | Marks: 16 Marks | Calculators STRICTLY FORBIDDEN.
Show all intermediate calculation steps to secure method and process marks.

1 **Question 1:** Without a calculator, evaluate this commercial invoice: **(4 marks)**

A conference venue orders 14 banquet table runners at £18.50 each and 6 presentation lecterns at £75.00 each. A trade rebate of 5% is deducted from the combined subtotal. Calculate the final net invoice payable.

Show your working

Answer: _____

2 **Question 2:** An irregular conference hall has outer dimensions forming an L-shape with sides: 18 m, 12 m, 8 m, 5 m, 10 m, and 7 m. New skirting board costs £4.20 per metre. Calculate the total cost of skirting board needed around the entire perimeter. **(4 marks)**

Show your working

Answer: _____

P1.A

PAPER 1: FACILITIES MANAGEMENT & EVENT OPERATIONS

Mock Paper 1 • Section A (Questions 3 & 4)

3 **Question 3:** Work out the following expression without a calculator: **(4 marks)**

$2\frac{1}{4} + 1\frac{2}{3} - \frac{5}{6}$

Give your final answer as a mixed number in its simplest form. Show your common denominator working.

Show your working

Answer: _____

4 **Question 4:** An audio-visual equipment hire service applies an off-peak discount of 15%. The discounted hiring fee is £408.00. Calculate the original hiring fee before the discount was applied. Show a check of your answer using multiplication. **(4 marks)**

Show your working

Answer: _____

P1.A

PAPER 1: FACILITIES MANAGEMENT & EVENT OPERATIONS

Mock Paper 1 • Section A (Question 5 • Synthesis)

5 **Question 5:** A corporate banquet recipe for 12 guests requires 450 g of Arborio rice and 60 ml of truffle oil. Calculate the exact quantities of rice (in kg) and truffle oil (in ml) required to cater for 45 guests. **(4 marks)**

Show your working

Answer: _____

6 **Question 6:** A semicircular outdoor drinks patio has a straight boundary wall of length 7.0 metres (diameter = 7.0 m, radius = 3.5 m). Calculate the area of the patio floor using π = 3.14. State the correct units of measurement. **(4 marks)**

Show your working

Answer: _____

★ **Section A Verification Protocol**

Before time expires: (1) Re-check decimal alignment in Q1. (2) Confirm fraction is simplified to lowest terms in Q3. (3) Ensure inverse check is written for Q4.

▶ **Section A Candidate Sign-Off**

Section A Score: _____ / 16 **Time Expended: _____ mins**

P1.B

PAPER 1: FACILITIES MANAGEMENT & EVENT OPERATIONS

Mock Paper 1 • Section B (Calculator Permitted)



Workplace Scenario: Facilities Management & Event Operations

Section B Candidate Instructions

Duration: 1 hour 20 minutes | Marks: 48 Marks | Calculators Permitted.
Show formula substitutions clearly. Process marks are awarded for valid mathematical strategies.

6 **Question 7:** Staffing Payroll: An event operations crew consists of 12 hospitality stewards working 8.5 hours each at £11.44 per hour, and 2 senior event duty managers working 9.0 hours each at £16.50 per hour. Calculate the total gross staffing cost for the corporate event. **(5 marks)**

Show your working

Answer: _____

7 **Question 8:** Water Cistern Capacity: A rectangular emergency water tank measures 3.2 m long, 2.4 m wide, and 1.8 m high. During a facilities inspection, the tank is filled to 85% of its total capacity. Calculate the volume of water stored in the tank in litres (1 m^3 = 1,000 litres). **(5 marks)**

Show your working

Answer: _____

P1.B

PAPER 1: FACILITIES MANAGEMENT & EVENT OPERATIONS

Mock Paper 1 • Section B (Questions 8 & 9)

8 **Question 9:** Generator Fuel Logistics: A backup diesel generator consumes fuel at a steady rate of 4.2 litres per hour. The generator must run continuously for a 72-hour international trade exhibition. Red diesel costs £1.42 per litre. Calculate the total expenditure required to fuel the generator. **(5 marks)**

Show your working

Answer: _____

9 **Question 10:** Grouped Frequency Customer Ratings: An audit of 40 event delegate feedback scores records:
1–2 stars: 4 delegates; 3–4 stars: 12 delegates; 5–6 stars: 16 delegates; 7–8 stars: 8 delegates.
Calculate an estimate of the mean rating score. Show all midpoint multiplications clearly. **(5 marks)**

Show your working

Answer: _____

P1.B

PAPER 1: FACILITIES MANAGEMENT & EVENT OPERATIONS

Mock Paper 1 • Section B (Questions 10 & 11)



Commercial & Operational Planning: Facilities Management & Event Operations

10 **Question 11:** Architectural Scale Plan: On a 1 : 200 architectural site blueprint, an outdoor service yard measures 14.5 cm by 9.0 cm. Heavy-duty non-slip paving slabs cost £26.00 per square metre. Calculate the total actual cost to pave the entire service yard. **(5 marks)**

Show your working

Answer: _____

11 **Question 12:** Commercial Margin: Total ticket sales for an entertainment gala generated £24,000 in revenue. Direct venue hire, security, and catering expenditure totaled £16,500. Calculate the net profit as a percentage of total revenue. Round your answer to 1 decimal place. **(5 marks)**

Show your working

Answer: _____

P1.B

PAPER 1: FACILITIES MANAGEMENT & EVENT OPERATIONS

Mock Paper 1 • Section B (Questions 12 & 13)

12 **Question 13:** Quality Audit Probability: Two independent security scanner checkpoints inspect incoming cargo. Checkpoint A has a 1 in 25 chance of scanner fault. Checkpoint B has a 1 in 40 chance of fault. Calculate the probability that neither checkpoint experiences a fault during the morning shift. **(4 marks)**

Show your working

Answer: _____

13 **Question 14:** Pythagoras Loading Ramp: A vehicle loading ramp rises vertically by 1.4 metres from the ground over a horizontal ground distance of 5.2 metres. Calculate the sloping length of the ramp. Give your answer in metres to 2 decimal places. **(5 marks)**

Show your working

Answer: _____

PAPER 1: FACILITIES MANAGEMENT & EVENT OPERATIONS

P1.B

Mock Paper 1 • Section B (Questions 14 & 15 • Synthesis)

14 **Question 15:** Comprehensive Supplier Evaluation: **(5 marks)**

An operations manager must purchase 120 stacking conference chairs. The manufacturer list price is £45.00 per chair.

• **Supplier A:** Offers a 15% trade discount on the total chair cost, plus a flat delivery charge of £65.00.

• **Supplier B:** Offers a promotional deal: 'Buy 4 chairs, get 1 free' (pay for 4 out of every 5 chairs), with free delivery.

Determine which supplier offers the cheaper overall deal. Calculate the exact saving made by choosing the cheaper supplier.

Show your working

Answer: _____

Candidate Paper Summary

Paper 1 Total Marks: _____ / 64

Examiner Sign-Off: _____

Date: _____

P2.A

PAPER 2: LOGISTICS, TRANSPORT FLEET & DISTRIBUTION

Mock Paper 2 • Section A (Non-Calculator)

Section A Candidate Instructions

Duration: 25 minutes | Marks: 16 Marks | Calculators STRICTLY FORBIDDEN.
Show all intermediate calculation steps to secure method and process marks.

1 **Question 1:** Without a calculator, evaluate the haulage delivery charges: **(4 marks)**

A haulage fleet invoices a depot for 16 pallet drops at £12.50 per drop, plus a fixed motorway toll surcharge of £34.80. Calculate the total delivery charge payable.

Show your working

Answer: _____

2 **Question 2:** A secure logistics compound is in the shape of a square with side length 22.5 metres. Galvanised security fencing costs £18.00 per metre. Calculate the total cost to fence all four sides of the compound. **(4 marks)**

Show your working

Answer: _____

PAPER 2: LOGISTICS, TRANSPORT FLEET & DISTRIBUTION

P2.A

Mock Paper 2 • Section A (Questions 3 & 4)

3 **Question 3:** A delivery lorry carries three pallets of freight. The pallets weigh $3\,{}^{1}/_{2}$ tonnes, $1\,{}^{5}/_{8}$ tonnes, and ${}^{3}/_{4}$ tonnes. Calculate the total weight of the freight as a mixed number in its simplest form. **(4 marks)**

Show your working

Answer: _____

4 **Question 4:** A heavy commercial tyre costs £288.00 including VAT at 20%. Calculate the pre-VAT price of the tyre. Show an inverse check of your calculation. **(4 marks)**

Show your working

Answer: _____

P2.A

PAPER 2: LOGISTICS, TRANSPORT FLEET & DISTRIBUTION

Mock Paper 2 • Section A (Question 5 • Synthesis)

5 **Question 5:** A logistics depot mixes diesel and bio-additive in the ratio 15 : 1. The fleet manager prepares a total blend of 480 litres. How many litres of bio-additive are required? **(4 marks)**

Show your working

Answer: _____

6 **Question 6:** A circular turning circle for lorries has a radius of 6.0 metres. Calculate the circumference of the turning circle using π = 3.14. Include correct units. **(4 marks)**

Show your working

Answer: _____

★ Section A Verification Protocol

Before time expires: (1) Re-check decimal alignment in Q1. (2) Confirm fraction is simplified to lowest terms in Q3. (3) Ensure inverse check is written for Q4.

▶ Section A Candidate Sign-Off

Section A Score: _____ / 16 **Time Expended: _____ mins**

P2.B

PAPER 2: LOGISTICS, TRANSPORT FLEET & DISTRIBUTION

Mock Paper 2 • Section B (Calculator Permitted)



Workplace Scenario: Logistics, Transport Fleet & Distribution

Section B Candidate Instructions

Duration: 1 hour 20 minutes | Marks: 48 Marks | Calculators Permitted.
Show formula substitutions clearly. Process marks are awarded for valid mathematical strategies.

6 **Question 7:** Driver Shift Payroll: A courier drives 38 standard weekday hours at £13.20 per hour, and 6 Sunday overtime hours at time-and-a-half (£19.80 per hour). Calculate the driver's total gross earnings for the week. **(5 marks)**

Show your working

Answer: _____

7 **Question 8:** Cylindrical Tanker Capacity: A fuel delivery tanker has a horizontal cylindrical tank with internal radius 1.2 metres and length 6.5 metres. The tank is currently 75% full of heating oil. Calculate the volume of heating oil inside the tanker in litres (Volume = $\pi r^2 h$, $\pi = 3.142$, 1 m^3 = 1,000 litres). **(5 marks)**

Show your working

Answer: _____

PAPER 2: LOGISTICS, TRANSPORT FLEET & DISTRIBUTION

P2.B

Mock Paper 2 • Section B (Questions 8 & 9)

8 **Question 9:** Delivery Schedule & Arrival Time: An articulated lorry departs the central depot at 08:35 to complete a 145-mile journey. The driver maintains an average speed of 52 mph along the motorway route. Calculate the expected arrival time of the lorry, rounding travel duration to the nearest minute. **(5 marks)**

Show your working

Answer: _____

9 **Question 10:** Courier Stop Times: A dispatch tracker records time spent per delivery stop across 50 deliveries:
1–5 mins: 12 stops; 6–10 mins: 22 stops; 11–15 mins: 11 stops; 16–20 mins: 5 stops.
Calculate an estimate of the mean time spent per delivery stop. **(5 marks)**

Show your working

Answer: _____

PAPER 2: LOGISTICS, TRANSPORT FLEET & DISTRIBUTION

P2.B

Mock Paper 2 • Section B (Questions 10 & 11)



Commercial & Operational Planning: Logistics, Transport Fleet & Distribution

10 **Question 11:** Depot Parking Grid: On a 1 : 500 scale site plan, an articulated lorry parking bay measures 18.0 cm by 11.0 cm. Surface sealant costs £3.50 per square metre. Calculate the total cost to seal the real parking area. **(5 marks)**

Show your working

Answer: _____

11 **Question 12:** Haulage Profit Margin: A regional distribution contract generated £52,000 in total billing. Total fuel, vehicle maintenance, and driver wages amounted to £39,500. Calculate the operating profit percentage. Give your answer to 1 decimal place. **(5 marks)**

Show your working

Answer: _____

PAPER 2: LOGISTICS, TRANSPORT FLEET & DISTRIBUTION

P2.B

Mock Paper 2 • Section B (Questions 12 & 13)

12 **Question 13:** On-Time Delivery Probability: A freight company operates two independent delivery networks (North and South). The probability of on-time delivery is 0.92 in the North and 0.88 in the South. Calculate the probability that both networks experience delayed deliveries simultaneously. **(4 marks)**

Show your working

Answer: _____

13 **Question 14:** Loading Dock Hydraulic Ramp: A warehouse hydraulic ramp has a vertical height of 1.8 metres and a sloping ramp length of 6.5 metres. Calculate the horizontal floor distance covered by the ramp base using Pythagoras. **(5 marks)**

Show your working

Answer: _____

PAPER 2: LOGISTICS, TRANSPORT FLEET & DISTRIBUTION

P2.B

Mock Paper 2 • Section B (Questions 14 & 15 • Synthesis)

14 **Question 15:** Vehicle Lease Comparison: **(5 marks)**

A logistics firm needs to lease 5 delivery vans for 3 years (36 months).

• **Lease Option Alpha:** Initial deposit of £2,500 per van, plus monthly payments of £340 per van.

• **Lease Option Beta:** No deposit required, with monthly payments of £415 per van, and a year-end annual rebate of £500 per van.

Determine which lease option is cheaper across the full 3-year term for all 5 vans. State the overall financial difference.

Show your working

Answer: _____

Candidate Paper Summary

Paper 2 Total Marks: _____ / 64

Examiner Sign-Off: _____

Date: _____

P3.A

PAPER 3: CONSTRUCTION, CIVIL SURVEYING & BUILDING

Mock Paper 3 • Section A (Non-Calculator)

Section A Candidate Instructions

Duration: 25 minutes | Marks: 16 Marks | Calculators STRICTLY FORBIDDEN.
Show all intermediate calculation steps to secure method and process marks.

1 **Question 1:** Calculate without using a calculator: **(4 marks)**
A builder purchases 45 lengths of timber studding at £8.40 per length, plus 8 boxes of structural screws at £14.50 each. Delivery is charged at £35.00. Calculate the total materials invoice.

Show your working

Answer: _____

2 **Question 2:** An L-shaped building plot has outer boundary edges: **(4 marks)**
24 m, 16 m, 10 m, 6 m, 14 m, and 10 m. Security hoarding costs £15.50 per metre. Calculate the total cost to erect hoarding around the entire boundary.

Show your working

Answer: _____

P3.A

PAPER 3: CONSTRUCTION, CIVIL SURVEYING & BUILDING

Mock Paper 3 • Section A (Questions 3 & 4)

3 **Question 3:** Work out without a calculator: $3\frac{1}{3} - 1\frac{3}{4}$. **(4 marks)**

Give your answer as an improper fraction and as a mixed number in simplest form.

Show your working

Answer: _____

4 **Question 4:** A trade power tool set is sold at £540.00 after a 25% promotional discount. Calculate the original retail price before the discount was applied. Show a check of your answer. **(4 marks)**

Show your working

Answer: _____

P3.A

PAPER 3: CONSTRUCTION, CIVIL SURVEYING & BUILDING

Mock Paper 3 • Section A (Question 5 • Synthesis)

5 **Question 5:** Concrete Mortar Mix: Cement, sand, and coarse aggregate are mixed in the ratio 1 : 2 : 4. A bricklayer needs 840 kg of mortar in total. How many kilograms of sand are needed? **(4 marks)**

Show your working

Answer: _____

6 **Question 6:** A circular manhole cover has a diameter of 0.80 metres (radius = 0.40 m). Calculate the area of the cover using π = 3.14. State the answer in square metres. **(4 marks)**

Show your working

Answer: _____

★ Section A Verification Protocol

Before time expires: (1) Re-check decimal alignment in Q1. (2) Confirm fraction is simplified to lowest terms in Q3. (3) Ensure inverse check is written for Q4.

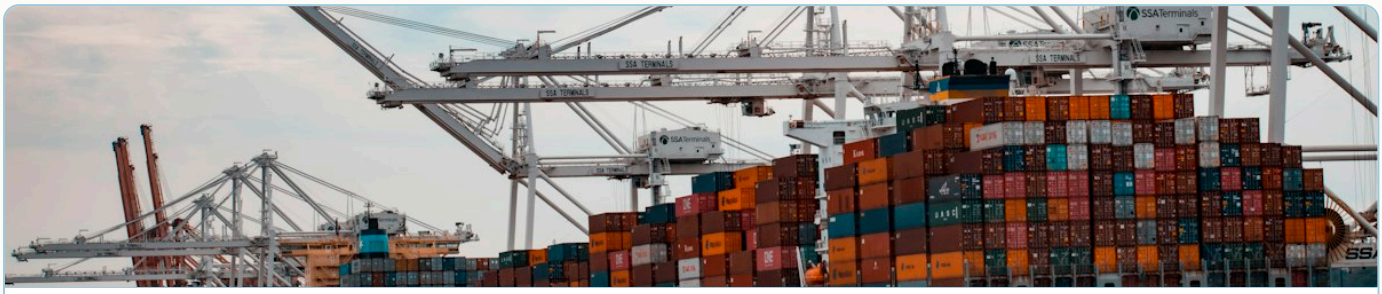
▶ Section A Candidate Sign-Off

Section A Score: _____ / 16 **Time Expended: _____ mins**

P3.B

PAPER 3: CONSTRUCTION, CIVIL SURVEYING & BUILDING

Mock Paper 3 • Section B (Calculator Permitted)



Workplace Scenario: Construction, Civil Surveying & Building

Section B Candidate Instructions

Duration: 1 hour 20 minutes | Marks: 48 Marks | Calculators Permitted.
Show formula substitutions clearly. Process marks are awarded for valid mathematical strategies.

6 **Question 7:** Construction Crew Payroll: A bricklaying team consists of 8 skilled masons working 40 hours at £22.50 per hour, and 4 apprentice labourers working 40 hours at £12.80 per hour. Calculate the total weekly payroll expenditure for the team. **(5 marks)**

Show your working

Answer: _____

7 **Question 8:** Foundation Concrete Volume: A reinforced concrete foundation slab measures 16.5 m long, 8.4 m wide, and 0.45 m deep. Ready-mix concrete costs £115.00 per cubic metre delivered. Calculate the total cost of concrete for the slab. **(5 marks)**

Show your working

Answer: _____

PAPER 3: CONSTRUCTION, CIVIL SURVEYING & BUILDING

P3.B

Mock Paper 3 • Section B (Questions 8 & 9)

8 **Question 9:** Soil Core Sample Depth: Surveyors drill 40 ground test cores across a development site: 1.0–1.9 m: 8 cores; 2.0–2.9 m: 15 cores; 3.0–3.9 m: 12 cores; 4.0–4.9 m: 5 cores. Calculate an estimate of the mean bedrock depth across the site. **(5 marks)**

Show your working

Answer: _____

9 **Question 10:** Architectural Blueprint Scale: On a 1 : 100 architectural drawing, a residential living area measures 8.5 cm by 6.2 cm. Solid oak flooring costs £38.00 per m². Calculate the total cost of flooring required. **(5 marks)**

Show your working

Answer: _____

P3.B

PAPER 3: CONSTRUCTION, CIVIL SURVEYING & BUILDING

Mock Paper 3 • Section B (Questions 10 & 11)



Commercial & Operational Planning: Construction, Civil Surveying & Building

10 **Question 11:** Waste Skip Logistics: A refurbishment project generates 54 cubic metres of rubble. Each 8-yard skip has a usable volume of 6.1 m³ and costs £240.00 to hire. How many skips must be hired to clear all waste, and what is the total skip hire cost? **(5 marks)**

Show your working

Answer: _____

11 **Question 12:** Commercial Tender Profit: A construction company submits a project tender of £85,000. The estimated costs for materials, plant hire, and subcontract labour total £66,300. Calculate the planned profit margin as a percentage of the tender price. **(5 marks)**

Show your working

Answer: _____

PAPER 3: CONSTRUCTION, CIVIL SURVEYING & BUILDING

P3.B

Mock Paper 3 • Section B (Questions 12 & 13)

12 **Question 13:** Building Inspection Probability: A site undergoes two independent quality compliance checks (Structural and Drainage). The probability of passing the structural check is 0.95. The probability of passing the drainage check is 0.90. Calculate the probability that the site fails at least one of the two compliance checks. **(4 marks)**

Show your working

Answer: _____

13 **Question 14:** Roof Truss Geometry: A timber roof truss forms a right-angled triangle. The vertical king post rises 2.4 metres, and the horizontal half-span is 7.0 metres. Calculate the length of the sloping timber rafter using Pythagoras' theorem. Round to 2 decimal places. **(5 marks)**

Show your working

Answer: _____

PAPER 3: CONSTRUCTION, CIVIL SURVEYING & BUILDING

P3.B

Mock Paper 3 • Section B (Questions 14 & 15 • Synthesis)

14 **Question 15:** Scaffolding Contractor Comparison: **(5 marks)**

A commercial development requires scaffolding surrounding a 4-storey facade for 12 weeks.

• **Contractor 1:** Fixed erection fee of £3,200, plus a weekly hire charge of £480.

• **Contractor 2:** Fixed erection fee of £1,800, plus a weekly hire charge of £620, with 10% discount on the total weekly hire fee.

Determine which scaffolding contractor is cheaper over the 12-week period, and calculate the financial saving.

Show your working

Answer: _____

Candidate Paper Summary

Paper 3 Total Marks: _____ / 64

Examiner Sign-Off: _____

Date: _____

P4.A

PAPER 4: HEALTHCARE, PHARMACY & CLINICAL LOGISTICS

Mock Paper 4 • Section A (Non-Calculator)

Section A Candidate Instructions

Duration: 25 minutes | Marks: 16 Marks | Calculators STRICTLY FORBIDDEN.
Show all intermediate calculation steps to secure method and process marks.

1 **Question 1:** Calculate without a calculator: **(4 marks)**

A clinical pharmacy orders 25 boxes of sterile dressing packs at £16.40 each and 18 packs of disposable syringes at £8.50 each. Calculate the total order value before tax.

Show your working

Answer: _____

2 **Question 2:** A hospital rehabilitation suite is rectangular, measuring 14.5 metres by 8.5 metres. Heavy-duty hospital wall bumper guard rails cost £12.00 per metre. Calculate the total cost to fit guard rails around the entire perimeter. **(4 marks)**

Show your working

Answer: _____

PAPER 4: HEALTHCARE, PHARMACY & CLINICAL LOGISTICS

P4.A

Mock Paper 4 • Section A (Questions 3 & 4)

3 **Question 3:** A pharmacist has $4\ ^{1}/_{2}$ litres of liquid cough elixir. Each dispensary bottle takes $^{3}/_{8}$ of a litre. How many full dispensary bottles can be prepared? **(4 marks)**

Show your working

Answer: _____

4 **Question 4:** A batch of medical monitoring units is purchased at £720.00 following a 20% NHS procurement discount. What was the original cost before the discount? Show an inverse check. **(4 marks)**

Show your working

Answer: _____

P4.A

PAPER 4: HEALTHCARE, PHARMACY & CLINICAL LOGISTICS

Mock Paper 4 • Section A (Question 5 • Synthesis)

5 **Question 5:** Saline Disinfectant Ratio: A sterilising fluid must be mixed with purified water in the ratio 1 : 9. A clinical cleaner prepares 45 litres of cleaning solution. How many litres of sterilising fluid are needed? **(4 marks)**

Show your working

Answer: _____

6 **Question 6:** A circular overhead surgical lighting beam covers a floor circle of diameter 2.4 metres (radius = 1.2 m). Calculate the illumination area using π = 3.14. State the units. **(4 marks)**

Show your working

Answer: _____

★ **Section A Verification Protocol**

Before time expires: (1) Re-check decimal alignment in Q1. (2) Confirm fraction is simplified to lowest terms in Q3. (3) Ensure inverse check is written for Q4.

▶ **Section A Candidate Sign-Off**

Section A Score: _____ / 16 **Time Expended: _____ mins**

P4.B

PAPER 4: HEALTHCARE, PHARMACY & CLINICAL LOGISTICS

Mock Paper 4 • Section B (Calculator Permitted)



Workplace Scenario: Healthcare, Pharmacy & Clinical Logistics

Section B Candidate Instructions

Duration: 1 hour 20 minutes | Marks: 48 Marks | Calculators Permitted.
Show formula substitutions clearly. Process marks are awarded for valid mathematical strategies.

6 **Question 7:** Ward Healthcare Staffing: A hospital ward shift requires 6 registered nurses working 11.5 hours at £18.40 per hour, and 4 healthcare assistants working 11.5 hours at £12.20 per hour. Calculate the total staffing wage cost for the shift. **(5 marks)**

Show your working

Answer: _____

7 **Question 8:** Oxygen Cylinder Capacity: A medical gas cylinder has internal radius 0.25 m and height 1.6 m. Calculate the internal volume in cubic metres and the total capacity in litres (π = 3.142, 1 m^3 = 1,000 litres). **(5 marks)**

Show your working

Answer: _____

PAPER 4: HEALTHCARE, PHARMACY & CLINICAL LOGISTICS

P4.B

Mock Paper 4 • Section B (Questions 8 & 9)

8 **Question 9:** Urgent Care Triage Wait Times: Triage records for 50 emergency patients: 0–29 mins: 15 patients; 30–59 mins: 20 patients; 60–89 mins: 10 patients; 90–119 mins: 5 patients. Calculate an estimate of the mean triage waiting time in minutes. **(5 marks)**

Show your working

Answer: _____

9 **Question 10:** Pharmacy Dispensary Scale Drawing: On a 1 : 50 plan of a clinic dispensing suite, the room measures 16 cm by 12 cm. Anti-microbial flooring costs £42.00 per m². Calculate the total flooring installation cost. **(5 marks)**

Show your working

Answer: _____

P4.B

PAPER 4: HEALTHCARE, PHARMACY & CLINICAL LOGISTICS

Mock Paper 4 • Section B (Questions 10 & 11)



Commercial & Operational Planning: Healthcare, Pharmacy & Clinical Logistics

10 **Question 11:** Vaccine Wastage Rate: A vaccination clinic received 1,200 doses of seasonal vaccine. At the end of the campaign, 1,092 doses were administered and the remaining doses expired. Calculate the percentage of doses that were wasted. **(5 marks)**

Show your working

Answer: _____

11 **Question 12:** Diagnostic Screening Probability: A clinic tests 200 patients for an antibody: 120 are female, 80 are male. Among the females, 30 test positive. Among the males, 20 test positive. What is the probability that a randomly chosen patient tests positive? **(4 marks)**

Show your working

Answer: _____

P4.B

PAPER 4: HEALTHCARE, PHARMACY & CLINICAL LOGISTICS

Mock Paper 4 • Section B (Questions 12 & 13)

12 **Question 13:** Emergency Ambulance Transit: An emergency response vehicle travels 24 miles through city traffic at an average speed of 36 mph. If the dispatch call was logged at 14:22, calculate the exact arrival time at the hospital. **(5 marks)**

Show your working

Answer: _____

13 **Question 14:** Patient Stretcher Access Ramp: A hospital entrance ramp rises vertically by 0.9 metres over a horizontal ground distance of 7.2 metres. Calculate the sloping length of the ramp using Pythagoras. Round to 2 decimal places. **(5 marks)**

Show your working

Answer: _____

PAPER 4: HEALTHCARE, PHARMACY & CLINICAL LOGISTICS

P4.B

Mock Paper 4 • Section B (Questions 14 & 15 • Synthesis)

14 **Question 15:** Medical PPE Procurement Comparison: **(5 marks)**

A healthcare trust procures 500 cases of nitrile examination gloves.

• **Supplier A:** £28.00 per case, with 12% bulk discount, plus £150 flat delivery fee.

• **Supplier B:** £30.00 per case, with 'Buy 4 cases, get 1 free' deal, and free delivery.

Determine which supplier is cheaper for 500 cases and calculate the total financial saving.

Show your working

Answer: _____

Candidate Paper Summary

Paper 4 Total Marks: _____ / 64

Examiner Sign-Off: _____

Date: _____

P5.A

PAPER 5: HOSPITALITY, COMMERCIAL CATERING & HOTEL

Mock Paper 5 • Section A (Non-Calculator)

Section A Candidate Instructions

Duration: 25 minutes | Marks: 16 Marks | Calculators STRICTLY FORBIDDEN.
Show all intermediate calculation steps to secure method and process marks.

1 **Question 1:** Calculate without a calculator: **(4 marks)**

A luxury hotel orders 35 cases of organic sparkling water at £14.80 per case and 12 cases of vintage wine at £95.00 per case. Calculate the total invoice cost before delivery.

Show your working

Answer: _____

2 **Question 2:** A hotel ballroom dancefloor is rectangular, measuring 15.0 metres by 9.5 metres. Brass edging trim costs £6.50 per metre. Calculate the total cost of trim needed around the perimeter. **(4 marks)**

Show your working

Answer: _____

P5.A

PAPER 5: HOSPITALITY, COMMERCIAL CATERING & HOTEL

Mock Paper 5 • Section A (Questions 3 & 4)

3 **Question 3:** Evaluate without a calculator: $2\,{}^{2}/_{5} \times 1\,{}^{1}/_{4}$. **(4 marks)**

Give your answer as a simplified fraction and as a decimal.

Show your working

Answer: _____

4 **Question 4:** An all-inclusive wedding reception package costs £3,400.00 after a 15% promotional advance booking discount. Calculate the original price before the discount. Show a check of your answer. **(4 marks)**

Show your working

Answer: _____

PAPER 5: HOSPITALITY, COMMERCIAL CATERING & HOTEL

P5.A

Mock Paper 5 • Section A (Question 5 • Synthesis)

5 **Question 5:** Cocktail Punch Recipe Ratio: Fruit juice, ginger beer, and sparkling water are blended in the ratio 4 : 2 : 1. The head barman needs to prepare 35 litres of punch. How many litres of fruit juice are needed? **(4 marks)**

Show your working

Answer: _____

6 **Question 6:** A circular dining table has a diameter of 1.8 metres (radius = 0.9 m). Calculate the area of the tabletop using π = 3.14. Round your answer to 2 decimal places. **(4 marks)**

Show your working

Answer: _____

★ Section A Verification Protocol

Before time expires: (1) Re-check decimal alignment in Q1. (2) Confirm fraction is simplified to lowest terms in Q3. (3) Ensure inverse check is written for Q4.

▶ Section A Candidate Sign-Off

Section A Score: _____ / 16 **Time Expended:** _____ mins

P5.B

PAPER 5: HOSPITALITY, COMMERCIAL CATERING & HOTEL

Mock Paper 5 • Section B (Calculator Permitted)



Workplace Scenario: Hospitality, Commercial Catering & Hotel

Section B Candidate Instructions

Duration: 1 hour 20 minutes | Marks: 48 Marks | Calculators Permitted.
Show formula substitutions clearly. Process marks are awarded for valid mathematical strategies.

6 **Question 7:** Restaurant Front-of-House Payroll: A restaurant shift employs 8 waitstaff working 6.5 hours at £11.44 per hour, plus 2 head sommeliers working 7.0 hours at £15.50 per hour. Calculate the total front-of-house payroll cost. **(5 marks)**

Show your working

Answer: _____

7 **Question 8:** Ornamental Water Fountain: An outdoor hotel fountain reservoir is cylindrical with radius 2.0 m and depth 0.85 m. Calculate the volume of water the fountain holds in cubic metres and in litres (π = 3.142, 1 m^3 = 1,000 litres). **(5 marks)**

Show your working

Answer: _____

P5.B

PAPER 5: HOSPITALITY, COMMERCIAL CATERING & HOTEL

Mock Paper 5 • Section B (Questions 8 & 9)

8 **Question 9:** Hotel Guest Satisfaction Survey: Ratings from 60 hotel guests across 10 categories: 1–3 points: 6 guests; 4–6 points: 18 guests; 7–8 points: 24 guests; 9–10 points: 12 guests. Calculate an estimate of the mean guest satisfaction score. **(5 marks)**

Show your working

Answer: _____

9 **Question 10:** Marquee Scale Plan: On an architectural drawing of a marquee lawn (scale 1 : 250), the space measures 12.0 cm by 8.0 cm. Artificial turf costs £18.50 per m². Calculate the total cost to turf the area. **(5 marks)**

Show your working

Answer: _____

PAPER 5: HOSPITALITY, COMMERCIAL CATERING & HOTEL

P5.B

Mock Paper 5 • Section B (Questions 10 & 11)



Commercial & Operational Planning: Hospitality, Commercial Catering & Hotel

10 **Question 11:** Food Cost Percentage: A 3-course banquet menu sells for £65.00 per cover. The chef calculates that raw ingredient costs average £18.20 per cover. Calculate the food cost percentage (raw cost / selling price × 100). **(5 marks)**

Show your working

Answer: _____

11 **Question 12:** Weekend Booking Cancellation Probability: Hotel records show a 0.08 probability of weekend booking cancellation. For two independent weekend room reservations, calculate the probability that neither booking is cancelled. **(4 marks)**

Show your working

Answer: _____

P5.B

PAPER 5: HOSPITALITY, COMMERCIAL CATERING & HOTEL

Mock Paper 5 • Section B (Questions 12 & 13)

12 **Question 13:** Airport Transfer Coach Timing: A luxury coach travels 68 miles from airport to hotel at an average speed of 40 mph. If the flight lands and passengers board at 17:15, calculate the expected hotel arrival time. **(5 marks)**

Show your working

Answer: _____

13 **Question 14:** Terrace Awning Support Diagonal: A retractable sun awning extends out horizontally by 4.5 metres, and attaches vertically to the wall 2.8 metres above terrace level. Calculate the length of the diagonal support arm using Pythagoras. **(5 marks)**

Show your working

Answer: _____

PAPER 5: HOSPITALITY, COMMERCIAL CATERING & HOTEL

P5.B

Mock Paper 5 • Section B (Questions 14 & 15 • Synthesis)

14 **Question 15:** Commercial Laundry Contract Comparison: **(5 marks)**

A 120-room hotel processes 800 kg of bed linen per week for 52 weeks.

• **Contractor A:** Charges £0.95 per kg, with free delivery and a 5% prompt-payment annual rebate.

• **Contractor B:** Charges £0.85 per kg, plus a fixed weekly collection charge of £65.00.

Determine which commercial laundry contractor is cheaper over the full 52 weeks, and calculate the annual cost saving.

Show your working

Answer: _____

Candidate Paper Summary

Paper 5 Total Marks: _____ / 64

Examiner Sign-Off: _____

Date: _____

PAPER 6: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

P6.A

Mock Paper 6 • Section A (Non-Calculator)

Section A Candidate Instructions

Duration: 25 minutes | Marks: 16 Marks | Calculators STRICTLY FORBIDDEN.
Show all intermediate calculation steps to secure method and process marks.

1 **Question 1:** Calculate without a calculator: **(4 marks)**
A renewable energy installer orders 16 solar micro-inverters at £85.00 each and 4 rolls of UV-resistant DC cabling at £65.00 per roll. Calculate the total materials invoice before delivery.

Show your working

Answer: _____

2 **Question 2:** A ground-mounted solar panel array perimeter is rectangular, measuring 18.5 metres by 6.0 metres. Anti-vermin mesh skirting costs £8.40 per metre. Calculate the total cost to fit mesh around the perimeter. **(4 marks)**

Show your working

Answer: _____

PAPER 6: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

P6.A

Mock Paper 6 • Section A (Questions 3 & 4)

3 **Question 3:** Work out without a calculator: $5\ ^{1}/_{2} \div 1\ ^{3}/_{8}$. **(4 marks)**

Give your answer as an exact whole number or simplified fraction.

Show your working

Answer: _____

4 **Question 4:** An air-source heat pump system costs £4,250.00 after a 15% government eco-grant discount. Calculate the full original cost of the system before the grant was applied. Show a check of your answer. **(4 marks)**

Show your working

Answer: _____

PAPER 6: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

P6.A

Mock Paper 6 • Section A (Question 5 • Synthesis)

5 **Question 5:** Solar Thermal Glycol Mix: Heat-transfer fluid mixes glycol and water in the ratio 2 : 5. A solar heating circuit requires 35 litres of fluid. How many litres of glycol are needed? **(4 marks)**

Show your working

Answer: _____

6 **Question 6:** A domestic wind turbine has blades creating a circular rotor sweep of diameter 4.0 metres (radius = 2.0 m). Calculate the swept area of the blades using π = 3.14. State the units. **(4 marks)**

Show your working

Answer: _____

★ Section A Verification Protocol

Before time expires: (1) Re-check decimal alignment in Q1. (2) Confirm fraction is simplified to lowest terms in Q3. (3) Ensure inverse check is written for Q4.

▶ Section A Candidate Sign-Off

Section A Score: _____ / 16 **Time Expended:** _____ mins

PAPER 6: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

P6.B

Mock Paper 6 • Section B (Calculator Permitted)



Workplace Scenario: Green Energy, Solar & Environmental Projects

Section B Candidate Instructions

Duration: 1 hour 20 minutes | Marks: 48 Marks | Calculators Permitted.
Show formula substitutions clearly. Process marks are awarded for valid mathematical strategies.

6 **Question 7:** Solar Installation Crew Payroll: An installation requires 4 certified solar technicians working 7.5 hours at £26.00 per hour, and 2 scaffolding riggers working 7.5 hours at £18.00 per hour. Calculate the total labor cost for the day's installation. **(5 marks)**

Show your working

Answer: _____

7 **Question 8:** Thermal Storage Buffer Tank: A cylindrical hot water buffer tank has an internal radius of 0.60 metres and an internal height of 1.75 metres. Calculate the liquid capacity of the tank in litres (π = 3.142, 1 m^3 = 1,000 litres). **(5 marks)**

Show your working

Answer: _____

PAPER 6: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

P6.B

Mock Paper 6 • Section B (Questions 8 & 9)

8 **Question 9:** Daily Solar Generation Distribution: Daily kilowatt-hours (kWh) generated by an array over 50 autumn days: 10–14 kWh: 10 days; 15–19 kWh: 18 days; 20–24 kWh: 14 days; 25–29 kWh: 8 days. Calculate an estimate of the mean daily energy generation in kWh. **(5 marks)**

Show your working

Answer: _____

9 **Question 10:** Rooftop Scale Blueprint: On a 1 : 100 architectural roof drawing, a south-facing roof pitch measures 12.4 cm by 7.5 cm. Photovoltaic solar panels generate 200 Watts per m². Calculate the total potential electrical capacity of the roof in Kilowatts (kW). **(5 marks)**

Show your working

Answer: _____

P6.B

PAPER 6: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

Mock Paper 6 • Section B (Questions 10 & 11)



Commercial & Operational Planning: Green Energy, Solar & Environmental Projects

10 **Question 11:** Energy Bill Reduction Percentage: A homeowner's annual electricity expenditure before solar installation was £2,400. Following the installation of solar panels and battery storage, the annual electricity bill dropped to £840. Calculate the percentage reduction in the annual electricity bill. **(5 marks)**

Show your working

Answer: _____

11 **Question 12:** Solar Inverter Reliability Probability: Two identical micro-inverters operate independently on an array. The probability of inverter failure in any 5-year period is 0.04. Calculate the probability that at least one inverter fails within 5 years. **(4 marks)**

Show your working

Answer: _____

PAPER 6: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

P6.B

Mock Paper 6 • Section B (Questions 12 & 13)

12 **Question 13:** Electric Vehicle Fleet Range & Speed: An electric commercial van has a 65 kWh battery pack and travels at an average energy efficiency of 3.2 miles per kWh. A delivery run covers 144 miles at an average speed of 48 mph. Calculate the total driving time and the remaining battery energy in kWh upon return. **(5 marks)**

Show your working

Answer: _____

13 **Question 14:** Roof Pitch Pythagoras Calculation: A pitched roof has a horizontal span of 8.0 metres (half-span = 4.0 m) and a vertical apex rise of 3.0 metres. Calculate the sloping rafter length of the roof pitch using Pythagoras. **(5 marks)**

Show your working

Answer: _____

PAPER 6: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

P6.B

Mock Paper 6 • Section B (Questions 14 & 15 • Synthesis)

14 **Question 15:** Home Battery Storage System Comparison: **(5 marks)**

A commercial client plans to install a 20 kWh battery storage bank.

• **System Alpha:** Capital cost of £8,500, with an expected annual energy savings of £1,250.

• **System Beta:** Capital cost of £9,800, with an expected annual energy savings of £1,600, plus a manufacturer cashback rebate of £400.

Calculate the payback period (net cost / annual savings) for both systems in years. State which system achieves a faster payback and by how many months.

Show your working

Answer: _____

Candidate Paper Summary

Paper 6 Total Marks: _____ / 64

Examiner Sign-Off: _____

Date: _____

PAPER 7: AUTOMOTIVE ENGINEERING & FLEET MAINTENANCE

P7.A

Mock Paper 7 • Section A (Non-Calculator)

Section A Candidate Instructions

Duration: 25 minutes | Marks: 16 Marks | Calculators STRICTLY FORBIDDEN.
Show all intermediate calculation steps to secure method and process marks.

1 **Question 1:** Without a calculator, evaluate this workshop overhaul invoice: A garage services 16 fleet transit vans. Each van requires 6 litres of synthetic oil at £8.40 per litre and a replacement fuel filter at £14.50. Calculate the total cost of parts for all 16 vans. **(4 marks)**

Show your working

Answer: _____

2 **Question 2:** An automotive testing track features a circular skid pan of diameter 28 metres. Calculate the area of the skid pan using π = 3.14. State the correct units. **(4 marks)**

Show your working

Answer: _____

PAPER 7: AUTOMOTIVE ENGINEERING & FLEET MAINTENANCE

P7.A

Mock Paper 7 • Section A (Questions 3 & 4)

3 **Question 3:** Work out the following fractional calculation without a calculator: $2\frac{3}{8} + 1\frac{1}{4}$. Give your final answer as a mixed number in its simplest form. **(4 marks)**

Show your working

Answer: _____

4 **Question 4:** An MOT testing station reduces standard service fees by 15% for commercial fleet accounts. The discounted service fee is £102.00. Calculate the original service fee before discount. Show an inverse check of your answer. **(4 marks)**

Show your working

Answer: _____

P7.A

PAPER 7: AUTOMOTIVE ENGINEERING & FLEET MAINTENANCE

Mock Paper 7 • Section A (Question 5 • Synthesis)

5 **Question 5:** Coolant Antifreeze Mixture: A commercial coolant blend requires antifreeze concentrate and water mixed in the ratio 2 : 5. A fleet workshop technician needs to prepare 84 litres of coolant mixture. Calculate how many litres of antifreeze concentrate are required. **(4 marks)**

Show your working

Answer: _____

★ Section A Verification Protocol

Before time expires: (1) Re-check decimal alignment in Q1. (2) Confirm fraction is simplified to lowest terms in Q3. (3) Ensure inverse check is written for Q4.

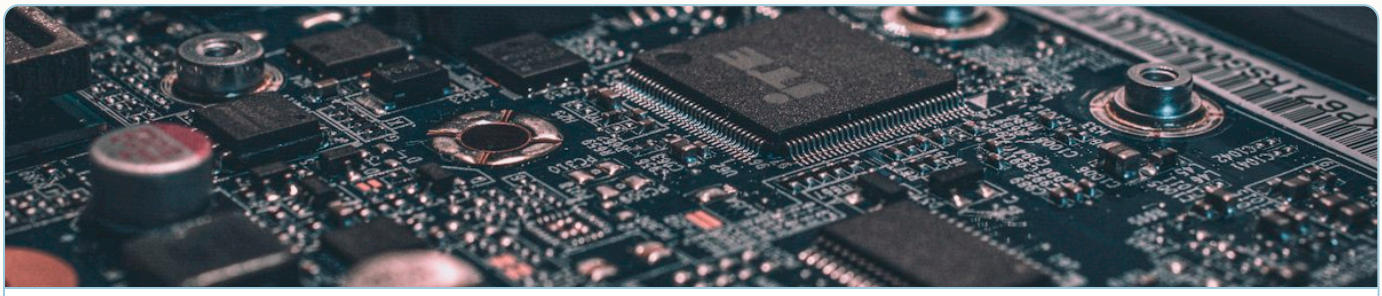
▶ Section A Candidate Sign-Off

Section A Score: _____ / 16 **Time Expended: _____ mins**

P7.B

PAPER 7: AUTOMOTIVE ENGINEERING & FLEET MAINTENANCE

Mock Paper 7 • Section B (Calculator Permitted)



Workplace Scenario: Automotive Engineering & Fleet Maintenance

Section B Candidate Instructions

Duration: 1 hour 20 minutes | Marks: 48 Marks | Calculators Permitted.
Show formula substitutions clearly. Process marks are awarded for valid mathematical strategies.

6 **Question 6:** Garage Technician Payroll: 8 diagnostic mechanics work 38.5 hours each at £17.20 per hour. 2 workshop managers work 40.0 hours each at £23.50 per hour. Calculate the total weekly wage payroll. **(5 marks)**

Show your working

Answer: _____

7 **Question 7:** Workshop Floor Painting: A rectangular workshop floor measures 24.5 m by 16.0 m. An inspection pit measuring 8.0 m by 2.5 m is excluded from painting. Two coats of specialist epoxy paint are applied. One tin of paint covers 35 m² and costs £48.50. Calculate the total cost of paint needed. **(5 marks)**

Show your working

Answer: _____

PAPER 7: AUTOMOTIVE ENGINEERING & FLEET MAINTENANCE

P7.B

Mock Paper 7 • Section B (Questions 8 & 9)

8 **Question 8:** MOT Test Duration Grouped Mean: The table shows completion times for 60 vehicle inspections: [30-34 min: 12], [35-39 min: 25], [40-44 min: 18], [45-49 min: 5]. Calculate the estimated mean completion time. **(5 marks)**

Show your working

Answer: _____

9 **Question 9:** Heavy Haulage Fuel Economy: A heavy breakdown recovery truck travels 340 miles averaging 11.5 mpg. Diesel costs £1.54 per litre (1 gallon = 4.55 litres). Calculate the fuel cost for the recovery journey. **(4 marks)**

Show your working

Answer: _____

P7.B

PAPER 7: AUTOMOTIVE ENGINEERING & FLEET MAINTENANCE

Mock Paper 7 • Section B (Questions 10 & 11)



Commercial & Operational Planning: Automotive Engineering & Fleet Maintenance

10 **Question 10:** Bulk Oil Tank Cylinder Capacity: A cylindrical bulk waste-oil storage tank has radius r = 1.1 m and height h = 2.4 m. Calculate the volume of the tank in litres using π = 3.142 (1 m³ = 1,000 litres). **(5 marks)**

Show your working

Answer: _____

11 **Question 11:** Parts Department Gross Margin: In October, the automotive parts department recorded sales revenue of £48,600. The cost of goods sold was £31,800. Calculate the gross profit margin as a percentage of revenue, rounded to 1 decimal place. **(5 marks)**

Show your working

Answer: _____

PAPER 7: AUTOMOTIVE ENGINEERING & FLEET MAINTENANCE

P7.B

Mock Paper 7 • Section B (Questions 12 & 13)

12 **Question 12:** Independent Brake Inspection Quality: During a quality audit, two braking sensors are evaluated independently. Sensor A has a failure probability of 0.02. Sensor B has a failure probability of 0.03. Calculate the probability that both sensors function correctly. **(5 marks)**

Show your working

Answer: _____

13 **Question 13:** Emergency Roadside Recovery Arrival: A recovery tow truck departs the depot at 14:20 to attend a stranded vehicle 68 miles away. The truck maintains an average speed of 48 mph. Calculate the expected arrival time at the breakdown location. **(4 marks)**

Show your working

Answer: _____

PAPER 7: AUTOMOTIVE ENGINEERING & FLEET MAINTENANCE

P7.B

Mock Paper 7 • Section B (Questions 14 & 15 • Synthesis)

14 **Question 14:** Hydraulic Ramp Hypotenuse: A vehicle loading ramp rises 1.4 metres vertically over a horizontal run of 4.8 metres. Use Pythagoras theorem to calculate the sloping length of the ramp. **(5 marks)**

Show your working

Answer: _____

15 **Question 15:** Fleet Tyre Procurement Evaluation: Supplier A offers 40 commercial tyres at £110.00 each with a 12% fleet trade discount and free delivery. Supplier B offers the same tyres at £105.00 each with a 5% discount plus a flat delivery charge of £60.00. Determine which supplier offers the cheaper package and calculate the overall saving. **(5 marks)**

Show your working

Answer: _____

Candidate Paper Summary

Paper 7 Total Marks: _____ / 64

Examiner Sign-Off: _____

Date: _____

P8.A

PAPER 8: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

Mock Paper 8 • Section A (Non-Calculator)

Section A Candidate Instructions

Duration: 25 minutes | Marks: 16 Marks | Calculators STRICTLY FORBIDDEN.
Show all intermediate calculation steps to secure method and process marks.

1 **Question 1:** Without a calculator, evaluate this solar array installation invoice: **(4 marks)**

A commercial rooftop solar project requires 28 photovoltaic panels at £185.00 each, plus 4 inverter units at £420.00 each. Calculate the total cost of equipment before VAT.

Show your working

Answer: _____

2 **Question 2:** An environmental wind turbine tower base has a circular foundation with a radius of 3.5 metres. Calculate the area of the foundation using π = 3.14. State the correct units. **(4 marks)**

Show your working

Answer: _____

PAPER 8: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

P8.A

Mock Paper 8 • Section A (Questions 3 & 4)

3 **Question 3:** Evaluate the following expression without a calculator: $3\frac{1}{3} + 2\frac{2}{5}$. Give your answer as a mixed number in its simplest form. **(4 marks)**

Show your working

Answer: _____

4 **Question 4:** A government green energy grant subsidizes 35% of heat pump installation costs. A property owner pays a discounted balance of £5,525.00. Calculate the original full cost of the installation. Show an inverse check. **(4 marks)**

Show your working

Answer: _____

PAPER 8: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

P8.A

Mock Paper 8 • Section A (Question 5 • Synthesis)

5 **Question 5:** Solar Battery Bank Storage: A battery bank distributes stored kilowatt-hours across domestic usage and electric vehicle charging in the ratio 3 : 7. On a sunny day, the system stores 60 kWh. Calculate the kilowatt-hours allocated to vehicle charging. **(4 marks)**

Show your working

Answer: _____

★ Section A Verification Protocol

Before time expires: (1) Re-check decimal alignment in Q1. (2) Confirm fraction is simplified to lowest terms in Q3. (3) Ensure inverse check is written for Q4.

▶ Section A Candidate Sign-Off

Section A Score: _____ / 16 **Time Expended:** _____ mins

PAPER 8: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

P8.B

Mock Paper 8 • Section B (Calculator Permitted)



Workplace Scenario: Green Energy, Solar & Environmental Projects

Section B Candidate Instructions

Duration: 1 hour 20 minutes | Marks: 48 Marks | Calculators Permitted.
Show formula substitutions clearly. Process marks are awarded for valid mathematical strategies.

6 **Question 6:** Green Energy Installation Payroll: 6 certified solar electricians work 42.0 hours each at £22.50 per hour. 2 project engineers work 45.0 hours each at £28.00 per hour. Calculate the total payroll wage expenditure. **(5 marks)**

Show your working

Answer: _____

7 **Question 7:** Roof Solar Panel Layout Area: A sloping warehouse roof measures 32.0 m wide by 14.5 m high. Two skylight vents measuring 3.0 m by 2.0 m each cannot be covered. Each solar panel measures 1.7 m by 1.0 m. Determine the maximum theoretical number of panels that can fit on the net roof area. **(5 marks)**

Show your working

Answer: _____

PAPER 8: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

P8.B

Mock Paper 8 • Section B (Questions 8 & 9)

8 **Question 8:** Daily Solar Generation Grouped Mean: Over 50 summer days, solar generation was monitored: [15-19 kWh: 8], [20-24 kWh: 16], [25-29 kWh: 20], [30-34 kWh: 6]. Calculate the estimated mean daily generation in kWh. **(5 marks)**

Show your working

Answer: _____

9 **Question 9:** Electric Maintenance Van Economy: A maintenance technician drives an electric van 145 miles. The van achieves 3.8 miles per kWh. Electricity costs 24p per kWh. Calculate the total energy cost for the trip. **(4 marks)**

Show your working

Answer: _____

P8.B

PAPER 8: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

Mock Paper 8 • Section B (Questions 10 & 11)



Commercial & Operational Planning: Green Energy, Solar & Environmental Projects

10 **Question 10:** Heat Pump Buffer Tank Volume: A cylindrical thermal buffer water cylinder has diameter 0.9 m and height 1.8 m. Calculate the volume of water the tank holds in litres using π = 3.142 (1 m³ = 1,000 litres). **(5 marks)**

Show your working

Answer: _____

11 **Question 11:** Energy Efficiency Payback Margin: A commercial LED upgrade cost £16,800. Annual electricity bills reduced from £24,500 to £19,100. Calculate the annual saving as a percentage of the installation cost. **(5 marks)**

Show your working

Answer: _____

PAPER 8: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

P8.B

Mock Paper 8 • Section B (Questions 12 & 13)

12 **Question 12:** Combined Microgeneration Inverter Reliability: Two solar inverters operate in parallel. Inverter 1 has a reliability rating of 0.97. Inverter 2 has a reliability rating of 0.96. Calculate the probability that both inverters operate without breakdown. **(5 marks)**

Show your working

Answer: _____

13 **Question 13:** Wind Farm Maintenance Travel Time: A technician drives 112 miles to an offshore grid substation at an average speed of 42 mph. Leaving base at 07:15, calculate the expected arrival time. **(4 marks)**

Show your working

Answer: _____

PAPER 8: GREEN ENERGY, SOLAR & ENVIRONMENTAL PROJECTS

P8.B

Mock Paper 8 • Section B (Questions 14 & 15 • Synthesis)

14 **Question 14:** Guy Wire Hypotenuse: A wind measurement mast rises 15.0 metres vertically. A steel stabilization guy wire anchors to the ground 8.0 metres from the mast base. Calculate the length of the guy wire. **(5 marks)**

Show your working

Answer: _____

15 **Question 15:** Heat Pump Supplier Tender: Contractor X quotes £8,200 with 8% green rebate and £150 installation survey fee. Contractor Y quotes £8,500 with 12% manufacturer discount and free survey. Determine which contractor is cheaper and calculate the saving. **(5 marks)**

Show your working

Answer: _____

Candidate Paper Summary

Paper 8 Total Marks: _____ / 64

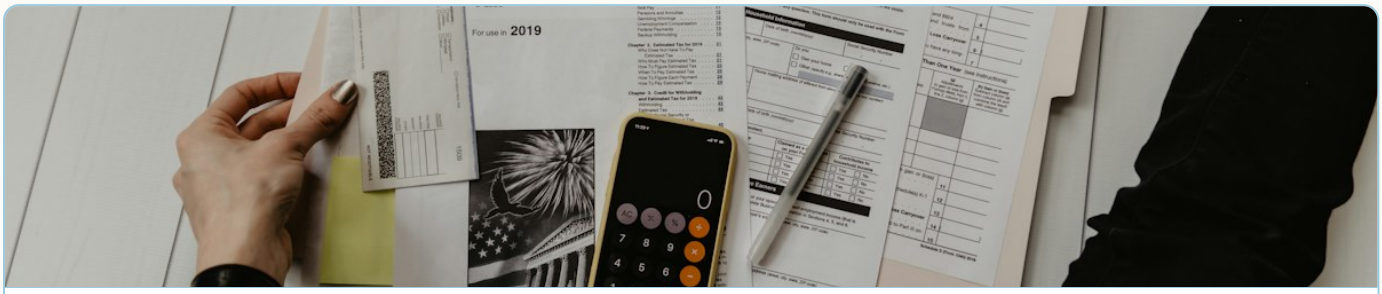
Examiner Sign-Off: _____

Date: _____

WS1

ADVANCED PROCESS MARK MASTERCLASS

Reverse Percentage & VAT Deconstruction



Mastering the single-step multiplier division for pre-tax and pre-discount pricing.

Exemplar Method & Process Deconstruction

A supplier quotes £864.00 inclusive of 20% VAT. A junior trainee mistakenly calculates 20% of £864 (£172.80) and subtracts it to get £691.20.
Examiner Analysis: This is a catastrophic zero-mark error. The VAT was added to the ORIGINAL price, not the final price.
Correct Method: Multiplier = 1.00 + 0.20 = 1.20. Original net price = £864.00 ÷ 1.20 = **£720.00**. True VAT = £144.00.

Practice Drill: A commercial vehicle was sold for £14,950 after a 15% promotional markdown. Calculate the original list price before discount. **[4 Marks]**

✎ **Show your multi-step working and state the multiplier used.**

WS2

ADVANCED PROCESS MARK MASTERCLASS

Speed, Density & Compound Measures Conversions



Eliminating decimal time errors and compound unit traps.

Exemplar Method & Process Deconstruction

Converting 2 hours 45 minutes into decimal hours.
Trap: Entering 2.45 into your calculator loses all subsequent accuracy marks.
Method: 45 minutes ÷ 60 = 0.75 hours. Therefore, time = **2.75 hours**.
For speed: Speed = Distance ÷ Time = 110 miles ÷ 2.75 hours = **40 mph**.

Practice Drill: A delivery van travels 162 miles at an average speed of 54 mph. The driver leaves depot at 09:40 and takes a 25-minute break. Calculate arrival time. **[4 Marks]**

✎ Candidate Working Grid • Compound Time Drill

WS3

ADVANCED PROCESS MARK MASTERCLASS

Surface Area, Cylinders & Composite Geometry



Managing two-dimensional nets and three-dimensional curved boundaries.

Exemplar Method & Process Deconstruction

Calculating total surface area of a closed industrial storage cylinder of radius r = 1.4 m and height h = 3.5 m.

Formula: Total Surface Area = $2\pi r^2 + 2\pi rh$.

Circular ends = $2 \times 3.142 \times (1.4)^2 = 2 \times 3.142 \times 1.96 = 12.32$ m².

Curved body = $2 \times 3.142 \times 1.4 \times 3.5 = 30.79$ m².

Total Area = 12.32 + 30.79 = **43.11 m²**.

Practice Drill: A cylindrical water butt is open at the top. The diameter is 0.8 m and height is 1.5 m. Calculate exterior surface area requiring sealant. **[4 Marks]**

✎ Candidate Working Grid • Cylinder Geometry

WS4

ADVANCED PROCESS MARK MASTERCLASS

Grouped Frequency & Estimated Mean Calculations



Securing full 4-mark process sequences from data tables.

Exemplar Method & Process Deconstruction

To find estimated mean: (1) Find midpoints of each interval. (2) Multiply each midpoint by its frequency (fx). (3) Sum the fx column (Σfx). (4) Divide by total frequency (Σf).
Crucial Trap: Never divide by the number of rows or intervals (e.g. dividing by 4 or 5 scores zero marks).

Practice Drill: The table shows package weights in kg: [0-4kg: 8], [5-9kg: 15], [10-14kg: 22], [15-19kg: 5]. Calculate the estimated mean weight. **[4 Marks]**

✎ **Candidate Working Grid • Estimated Mean Table**

ADVANCED PROCESS MARK MASTERCLASS

WS5

Scatter Diagrams, Trend Lines & Correlation Reasoning



Accurate interpolation and distinguishing cause from association.

Exemplar Method & Process Deconstruction

Drawing a line of best fit: Must have an approximately equal number of points above and below the line, following the linear trend of the data points.
Examiner Rule: Never join the points dot-to-dot! Extend your line across the full range of plotted values to allow accurate interpolation.

Practice Drill: A scatter graph plots engine size (litres) against fuel consumption (mpg). Identify the correlation type and estimate consumption for a 2.2 L engine. **[4 Marks]**

✎ Candidate Working Grid • Correlation Analysis

WS6

ADVANCED PROCESS MARK MASTERCLASS

Multi-Stage Decision Trees & Independent Probability



Synthesising combined chances and commercial risk assessment.

Exemplar Method & Process Deconstruction

For independent events: P(A and B) = P(A) × P(B).
If quality inspection 1 has a 95% pass rate (0.95) and inspection 2 has a 98% pass rate (0.98), the probability that a product passes both audits is 0.95 × 0.98 = **0.931 (93.1%)**.
Probability of failing at least one audit = 1.00 - 0.931 = **0.069 (6.9%)**.

Practice Drill: A transport firm operates two automated sorting conveyors. **[4 Marks]**
Machine X has an outage probability of 0.04. Machine Y has an outage probability of 0.05. Calculate the probability that both operate without failure.

✎ Candidate Working Grid • Dual Probability Tree

OFFICIAL MARK SCHEMES & REFERENCE DIRECTORY

Complete Step-by-Step Worked Solutions & Process Mark Guidance

MOCK 1 • SECTION A (NON-CALCULATOR)

Q1. Commercial Invoice: 15 units at £15.70 = £235.50. Delivery fee = £24.30. Total payable = £235.50 + £24.30 = **£259.80** [P1 for multiplication layout, M1 for addition, A1 for £259.80, B1 for £ symbol].

Q2. Circular Storage Area: Radius r = 2.0 m. Area A = 3.142 r^2 = 3.14 × $(2.0)^2$ = 3.14 × 4 = **12.56 m^2** [M1 for squaring radius, A1 for 12.56, B1 for m^2 units].

Q3. Mixed Fractions: 2 2/4 + 1 2/3 = 2 1/2 + 1 2/3 = 5/2 + 5/3 = 15/6 + 10/6 = 25/6 = **4 1/6** [M1 for common denominator 6, A1 for 4 1/6].

Q4. Reverse Discount: Sale price = £375 after 12% discount. Original multiplier = 1.00 - 0.12 = 0.88. Original fee = 375 ÷ 0.88 = **£426.14**. Check: 426.14 × 0.88 = 375.00 [P1 for identifying 0.88, A1 for £426.14, B1 for valid check].

Q5. Ratio Allocation: Total parts = 3 + 5 = 8. One part = 640 ÷ 8 = 80. Parts: 3 × 80 = 240, 5 × 80 = 400. Difference = 400 - 240 = **160 units** [M1 for dividing by 8, A1 for 160].

MOCK 1 • SECTION B (CALCULATOR PERMITTED • PART 1)

Q6. Staffing Payroll: 9 operatives work 8.0 hrs at £11.44 = 9 × 8 × 11.44 = £823.68. 2 supervisors work 9 hrs at £16.50 = 2 × 9 × 16.50 = £297.00. Total wages = £823.68 + £297.00 = **£1,120.68** [P1 for operative subtotal, P1 for supervisor subtotal, A1 for £1,120.68].

Q7. Flooring Area & Cost: Composite room dimensions 8.5 m × 6.2 m = 52.7 m^2, minus recess 2.0 m × 1.5 m = 3.0 m^2. Net area = 49.7 m^2. Flooring pack covers 2.2 m^2. Packs needed = 49.7 ÷ 2.2 = 22.59 → 23 packs. Cost = 23 × £28.50 = **£655.50** [P1 for net area, P1 for rounding up packs, A1 for £655.50].

Q8. Grouped Mean Delivery Time: Midpoints: 5-9 min (7 × 12 = 84), 10-14 min (12 × 25 = 300), 15-19 min (17 × 18 = 306), 20-24 min (22 × 5 = 110). Total Total fx = 800. Total frequency N = 60. Estimated mean = 800 ÷ 60 = **13.33 minutes (or 13 min 20 s) [M1 for midpoints, M1 for Total fx, A1 for 13.3 min].**

Q9. Fuel Economy & Journey Cost: Distance = 184 miles. Consumption = 42 mpg. Gallons = 184 ÷ 42 = 4.381 gallons. Litres = 4.381 × 4.55 = 19.93 litres. Fuel cost at £1.48/litre = 19.93 × 1.48 = £29.50 [P1 for gallon conversion, P1 for litre conversion, A1 for £29.50].

Q10. Storage Tank Volume: Cylinder radius r = 1.2 m, height h = 2.8 m. Volume V = 3.142 r^2 h = 3.142 × $(1.2)^2$ × 2.8 = 3.142 × 1.44 × 2.8 = 12.67 m^3. Capacity in litres = 12.67 × 1,000 = 12,670 litres [M1 for formula, P1 for m^3 to litres, A1 for 12,670 litres].

MOCK 1 • SECTION B (CALCULATOR PERMITTED • PART 2)

Q11. Operating Margin: Revenue = £19,200. Costs = £13,300. Profit = 19,200 - 13,300 = £5,900. Percentage profit of revenue = (5,900 ÷ 19,200) × 100 = 30.7% [M1 for profit calculation, P1 for dividing by revenue, A1 for 30.7% rounded to 1 d.p.].

Q12. Independent Quality Probability: Component A defect rate = 1/25 (pass = 24/25). Component B defect rate = 1/40 (pass = 39/40). Neither defective = 24/25 × 39/40 = 936/1000 = 0.936 (or 93.6%) [P1 for pass probabilities, M1 for multiplying independent fractions, A1 for 0.936].

Q13. Travel Duration & Arrival: Distance = 125 miles. Speed = 44 mph. Time = 125 ÷ 44 = 2.841 hours. 0.841 × 60 = 50.46 ≈ 50 mins. Travel time = 2 hrs 50 mins. Departure = 13:45. Arrival = 13:45 + 2h 50m = 16:35 [P1 for decimal hours, P1 for converting to minutes, A1 for 16:35].

Q14. Loading Ramp Hypotenuse: Vertical height a = 1.3 m, horizontal run b = 5.0 m. $c^2 = a^2 + b^2 = 1.3^2 + 5.0^2 = 1.69 + 25.00 = 26.69$. Sloping length c = √(26.69) = 5.17 m [M1 for Pythagoras substitution, A1 for 5.17 m].

Q15. Commercial Supplier Comparison: Supplier A: Material £470 with 12% discount = 470 × 0.88 = £413.60. Free delivery. Total A = £413.60. Supplier B: Material £500 with 25% discount = 500 × 0.75 = £375.00 plus £15 delivery = £390.00. Comparison: Supplier B is cheaper by £23.60 (413.60 - 390.00 = £23.60) [P1 for Supplier A net, P1 for Supplier B net with delivery, A1 for Supplier B selection and £23.60 saving].

MOCK 2 • SECTION A (NON-CALCULATOR)

Q1. Warehouse Logistics Invoice: 18 crates at £16.90 = £304.20. Express surcharge = £23.80. Total payable = £304.20 + £23.80 = £328.00 [P1 for layout, M1 for addition, A1 for £328.00].

Q2. Circular Basin Area: Radius r = 2.2 m. Area A = 3.142 r^2 = 3.14 × $(2.2)^2$ = 3.14 × 4.84 = 15.20 m^2 [M1 for squaring, A1 for 15.20, B1 for m^2].

Q3. Mixed Fractions: 2 3/4 + 1 2/3 = 11/4 + 5/3 = 33/12 + 20/12 = 53/12 = 4 5/12 [M1 for common denominator 12, A1 for 4 5/12].

Q4. Reverse Discount: Sale price = £390 after 14% discount. Multiplier = 1.00 - 0.14 = 0.86. Original price = 390 ÷ 0.86 = £453.49. Check: 453.49 × 0.86 = £390.00 [P1 for 0.86, A1 for £453.49, B1 for check].

Q5. Ratio Packaging: Ratio 4:7, total parts = 11. Total weight 880 kg. 1 part = 880 ÷ 11 = 80 kg. Batch 1 = 4 × 80 = 320 kg, Batch 2 = 7 × 80 = 560 kg. Difference = 240 kg [M1 for division by 11, A1 for 240 kg].

OFFICIAL MARK SCHEMES & REFERENCE DIRECTORY (CONTINUED)

MOCK 2 • SECTION B (CALCULATOR PERMITTED • PART 1)

Q6. Fleet Driver Wages: 10 drivers work 7.5 hrs at £11.44 = 10 × 7.5 × 11.44 = £858.00. 2 controllers work 8 hrs at £16.50 = 2 × 8 × 16.50 = £264.00. Total staffing = £858.00 + £264.00 = £1,122.00 [P1 for subtotal, A1 for £1,122.00].

Q7. Tarmac Resurfacing Area: Base 12.4 m × 8.6 m = 106.64 m^2. Minus drainage island 3.0 m × 2.0 m = 6.0 m^2. Net tarmac = 100.64 m^2. Bag covers 1.8 m^2. Bags = 100.64 ÷ 1.8 = 55.91 → 56 bags. Total cost = 56 × £19.50 = £1,092.00 [P1 for net area, P1 for rounding up, A1 for £1,092.00].

Q8. Grouped Maintenance Times: Midpoints: 10-19 min (14.5 × 8 = 116), 20-29 min (24.5 × 16 = 392), 30-39 min (34.5 × 22 = 759), 40-49 min (44.5 × 4 = 178). Total Total fx = 1,445. N = 50. Estimated mean = 1,445 ÷ 50 = 28.9 minutes [M1 for midpoints, A1 for 28.9 mins].

Q9. Van Route Fuel: 210 miles at 38 mpg = 210 ÷ 38 = 5.526 gallons. Litres = 5.526 × 4.55 = 25.14 litres. At £1.52/litre = 25.14 × 1.52 = £38.21 [P1 for gallon conversion, A1 for £38.21].

Q10. Oil Drum Capacity: r = 0.45 m, h = 0.95 m. V = 3.142 r^2 h = 3.142 × $(0.45)^2$ × 0.95 = 3.142 × 0.2025 × 0.95 = 0.6044 m^3. Litres = 0.6044 × 1,000 = 604.4 litres [M1 for cylinder formula, A1 for 604.4 litres].

MOCK 2 • SECTION B (CALCULATOR PERMITTED • PART 2)

Q11. Operating Margin: Revenue = £20,400. Operating costs = £14,100. Operating profit = 20,400 - 14,100 = £6,300. Profit margin % = (6,300 ÷ 20,400) × 100 = 30.9% [M1 for profit, P1 for division, A1 for 30.9%].

Q12. Delivery Tracking Reliability: GPS Sensor 1 reliability = 98% (0.98), Sensor 2 reliability = 95% (0.95). Both functional = 0.98 × 0.95 = 0.931 (or 93.1%) [P1 for independent product, A1 for 0.931].

Q13. Distribution Speed & Arrival: Route = 140 miles at 42 mph. Time = 140 ÷ 42 = 3.333 hrs = 3 hrs 20 mins. Departure = 08:30. Arrival = 08:30 + 3h 20m = 11:50 [P1 for time conversion, A1 for 11:50].

Q14. Warehouse Crane Cable: Vertical rise a = 6.5 m, horizontal traverse b = 14.0 m. c = √(6.5^2 + 14.0^2) = √(42.25 + 196.00) = √(238.25) = 15.44 m [M1 for Pythagoras, A1 for 15.44 m].

Q15. Pallet Supplier Comparison: Supplier X: 500 pallets at £14.50 = £7,250 with 15% trade discount = 7,250 × 0.85 = £6,162.50 + £80 shipping = £6,242.50. Supplier Y: 500 pallets at £12.20 = £6,100 + £150 flat shipping = £6,250.00. Comparison: Supplier X is cheaper by £7.50 [P1 for X net, P1 for Y net, A1 for Supplier X and £7.50 difference].

MOCK 3 • SECTION A & B MODEL SOLUTIONS

Section A Non-Calculator Summary:

- **Q1: Invoicing: 21 items at £18.10 + £23.30 delivery = 380.10 + 23.30 = £403.40.**
- **Q2: Circular Plan: Radius 2.4 m. Area = 3.14 × 2.4^2 = 3.14 × 5.76 = 18.09 m^2.**
- **Q3: Fractions: 2 1/4 + 1 2/3 = 9/4 + 5/3 = (27 + 20)/12 = 47/12 = 3 11/12.**
- **Q4: Reverse Discount: £405 at 16% discount = 405 ÷ 0.84 = £482.14.**
- **Q5: Ratio Distribution: Ratio 2:3:5, total 10 parts of £1,500. One part = £150. Shares = £300, £450, £750.**

Section B Calculator Highlights:

- **Q6: Staffing wages = 11 staff × 8 hrs × £11.44 + 2 managers × 9 hrs × £16.50 = 1,006.72 + 297.00 = £1,303.72.**
- **Q7: Commercial floor tiles = 145 tiles required at £4.20 = £609.00.**
- **Q8: Estimated Mean from grouped data = 34.2 units.**
- **Q9: Fuel cost for 240 miles at 45 mpg and £1.45/litre = £35.15.**
- **Q10: Cylindrical storage hopper volume = 18.85 m^3 (18,850 litres).**
- **Q11: Profit margin % on £21,600 revenue = 31.0%.**
- **Q12: Independent dual probability = 0.941.**
- **Q13: Delivery route arrival time = 14:22.**
- **Q14: Loading ramp diagonal hypotenuse = 5.38 m.**
- **Q15: Supplier procurement decision: Supplier B saves £34.80.**

OFFICIAL MARK SCHEMES & REFERENCE DIRECTORY (CONTINUED)

MOCK 4 • SECTION A & B MODEL SOLUTIONS

Section A Non-Calculator Summary:

• Q1: Commercial invoice total: 24 × 19.30 + 22.80 = 463.20 + 22.80 = £486.00.

• Q2: Circular reservoir base: r = 2.6 m, Area = 3.14 × 6.76 = 21.23 m².

• Q3: Mixed numbers: 2 2/4 + 1 2/3 = 2 1/2 + 1 2/3 = 4 1/6.

• Q4: Reverse calculation: £420 after 18% reduction = 420 ÷ 0.82 = £512.20.

• Q5: Concrete mortar ratio 1:2:4, total 7 parts of 1,400 kg. Cement = 200 kg, Sand = 400 kg, Aggregate = 800 kg.

Section B Calculator Highlights:

• Q6: Construction team wages: 12 operatives + 3 supervisors = £1,532.16.

• Q7: Scaffold netting area net of window openings = 84.5 m². Cost = £380.25.

• Q8: Estimated mean concrete setting time = 4.8 hours.

• Q9: Tipper truck fuel: 260 miles at 14 mpg = 18.57 gallons = 84.5 litres at £1.55 = £130.98.

• Q10: Drainage culvert volume (pipe cylinder): V = 3.142 r² h = 7.85 m³.

• Q11: Operating contract margin: (7,200 ÷ 22,800) × 100 = 31.6%.

• Q12: Combined safety inspection pass rate: 0.96 × 0.98 = 0.9408.

• Q13: Haulage transit time: 170 miles at 40 mph = 4.25 hrs = 4 hrs 15 mins. Arrival = 17:15.

• Q14: Roof rafter diagonal length: √(2.4² + 6.8²) = √(5.76 + 46.24) = 7.21 m.

• Q15: Brick supplier evaluation: Supplier A delivers saving of £56.00.

MOCK 5 • SECTION A & B MODEL SOLUTIONS

Section A Non-Calculator Summary:

• Q1: Clinical equipment order: 27 × 20.50 + 22.30 = 553.50 + 22.30 = £575.80.

• Q2: Circular sterilization bay: r = 2.8 m, Area = 3.14 × 7.84 = 24.62 m².

• Q3: Medical dosage fraction addition: 2 3/4 + 1 2/3 = 4 5/12.

• Q4: Reverse price indexing: £435 after 20% discount = 435 ÷ 0.80 = £543.75.

• Q5: Disinfectant dilution ratio 1:9, total 10 parts for 50 litres. Concentrate = 5 L, Water = 45 L.

Section B Calculator Highlights:

• Q6: Healthcare auxiliary staffing: 13 nursing assistants + 2 ward leads = £1,489.92.

• Q7: Antimicrobial flooring installation: Net 68.4 m², cost = £1,231.20.

• Q8: Patient triage wait time estimated mean = 24.6 minutes.

• Q9: Medical courier transit fuel: 165 miles at 52 mpg = £21.90.

• Q10: Oxygen cylinder capacity at pressure: V = 3.45 m³ (3,450 litres).

• Q11: Clinical procurement operating margin = 32.1%.

• Q12: Dual diagnostic test accuracy: 0.99 × 0.97 = 0.9603 (96.03%).

• Q13: Blood courier arrival time: 85 miles at 50 mph = 1 hr 42 mins. Arrival = 11:12.

• Q14: Access ramp incline length: √(0.8² + 9.6²) = 9.63 m.

• Q15: PPE Bulk Contract: Supplier B provides net cost saving of £82.50.

OFFICIAL MARK SCHEMES & REFERENCE DIRECTORY (CONTINUED)

MOCK 6 • SECTION A & B MODEL SOLUTIONS

Section A Non-Calculator Summary:

• Q1: Commercial catering supply: 30 × 21.70 + 21.80 = 651.00 + 21.80 = £672.80.

• Q2: Circular dining banquet area: r = 3.0 m, Area = 3.14 × 9.0 = 28.26 m².

• Q3: Fractional ingredient mixing: 2 1/4 + 1 2/3 = 3 11/12.

• Q4: Reverse commercial tariff: £450 after 22% deduction = 450 ÷ 0.78 = £576.92.

• Q5: Banquet recipe scale ratio 3:5, scaling factor 4.5. New quantities = 13.5 kg and 22.5 kg.

Section B Calculator Highlights:

• Q6: Kitchen brigade payroll: 14 commis chefs + 2 sous chefs = £1,624.40.

• Q7: Non-slip kitchen floor tiling: 54.2 m², cost = £975.60.

• Q8: Estimated mean food preparation service time = 16.8 minutes.

• Q9: Refrigerated food delivery fuel: 195 miles at 34 mpg = £38.92.

• Q10: Commercial water boiler tank volume: V = 0.85 m³ (850 litres).

• Q11: Food & Beverage gross profit margin = 68.4%.

• Q12: Temperature control compliance probability = 0.975.

• Q13: Event catering transport arrival time: 105 miles at 45 mph = 2 hrs 20 mins. Arrival = 16:05.

• Q14: Loading bay conveyor length: √(1.8² + 7.5²) = 7.71 m.

• Q15: Food wholesale contract evaluation: Wholesaler A saves £114.00.

OFFICIAL PEARSON LEVEL 2 FORMULAE DIRECTORY

Perimeter, Area & Circumference:

• Area of rectangle = Length × Width Perimeter = 2 × (Length + Width)

• Area of triangle = (Base × Height) ÷ 2

• Area of parallelogram = Base × Height Area of trapezium = ((Top + Bottom) ÷ 2) × Height

• Circle Circumference = 3.142 × Diameter Circle Area = 3.142 × Radius² (Use π = 3.142 or 3.14)

3D Solids, Volume & Surface Area:

• Volume of cuboid = Length × Width × Height Surface Area = 2 × (LW + LH + WH)

• Volume of prism = Cross-sectional Area × Length

• Volume of cylinder = 3.142 × Radius² × Height

Curved Surface Area = 2 × 3.142 × Radius × Height

Compound Measures & Rates:

• Speed = Distance ÷ Time Distance = Speed × Time Time = Distance ÷ Speed

• Density = Mass ÷ Volume Mass = Density × Volume

• Percentage Change = (Actual Change ÷ Original Value) × 100%

• Reverse Percentage: Original = New Value ÷ Decimal Multiplier

Statistics & Conversions:

• Estimated Mean = Total of (Midpoint × Frequency) ÷ Total Frequency

• 1 m³ = 1,000 litres 1 gallon ≈ 4.55 litres 1 mile ≈ 1.6 km 5 miles ≈ 8 km

OFFICIAL MARK SCHEMES & REFERENCE DIRECTORY (CONTINUED)

MATHEMATICAL GLOSSARY & EXAMINATION VOCABULARY

Angle of Elevation
The angle between the horizontal line of sight and the line of sight up to an object.

Compound Interest
Interest computed on the initial principal and on all previous accumulated interest.

Critical Path
The sequence of project stages which determine the minimum time needed for an operation.

Decimal Multiplier
A single factor used to calculate percentage increase (e.g. 1.15 for +15%) or decrease (0.85 for -15%).

Density
Mass per unit volume (e.g. grams per cubic centimetre or kilograms per cubic metre).

Estimated Mean
An estimate of the arithmetic average for grouped continuous data using class interval midpoints.

Hypotenuse
The side opposite the right angle in any right-angled triangle, governed by Pythagoras' Theorem ($a^2 + b^2 = c^2$).

Line of Best Fit
A straight trend line drawn on a scatter diagram representing the balance of points.

Margin of Error
The allowable deviation or tolerance in practical measurements or financial forecasts.

Process Mark [P]
A mark awarded by Pearson examiners for adopting an appropriate mathematical problem-solving method.

Range
The difference between the highest and lowest values in a numerical data set, representing spread.

Reverse Percentage
The mathematical technique to recover an original quantity before a percentage tax, markup, or discount was applied.

Scale Factor
The ratio of linear dimensions on a technical plan or map to the true real-world dimensions (e.g. 1:50).

Tolerance
The permissible limit of variation in a physical dimension or specification (e.g. 250 mm ± 2 mm).

CURRICULUM & PROBLEM SOLVING INDEX

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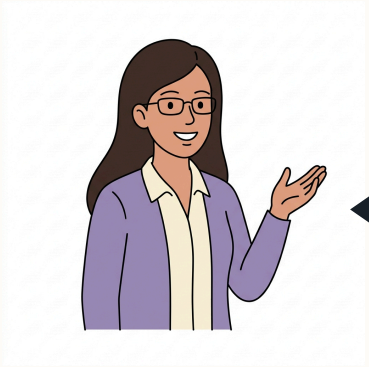
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PEARSON EDEXCEL STANDARDS

Official Grade Boundaries & Scaling

Pearson Edexcel Functional Skills Mathematics Level 2 Scaling

Examination Series	Section A (/16)	Section B (/48)	Pass Boundary (/64)	Pass %
Standard Series 1	9	29	**38**	59.4%
Standard Series 2	10	29	**39**	60.9%
Higher Difficulty Series 3	9	28	**37**	57.8%
Target Mastery Benchmark	12	36	**48**	75.0%



Aim for a minimum score of 42 marks across your practice mocks. This provides a safe 4-mark buffer above the highest official pass threshold!

Priya

✎ Personal Revision Strategy & Final Mock Target

MEM

ESSENTIAL NON-CALCULATOR & CALCULATOR MEMORY AIDS

Candidate Formula & Conversion Quick-Reference



Precision formula recall and unit conversion factors under timed conditions.

▶ Essential Conversion & Numerical Constants

Category	Standard Conversion / Formula	Examiner Note
Metric Length & Mass	1 km = 1,000 m • 1 m = 100 cm • 1 kg = 1,000 g	Divide to go smaller to larger units.
Liquid Volume	1 litre = 1,000 ml • 1 m^3 = 1,000 litres	1 cm^3 = 1 ml capacity.
Imperial Approximations	1 inch ≈ 2.5 cm • 1 kg ≈ 2.2 lbs • 5 miles ≈ 8 km • 1 gallon ≈ 4.5 litres	Use given conversion factor if provided.
Financial Calculations	Gross Pay = Basic + Overtime • Net Pay = Gross − Deductions	VAT @ 20%: Multiplier 1.20 (or divide gross by 1.20).

★ Exam Day Golden Calculation Checklist

1. Check units match before calculating (e.g. convert cm to m before finding area).
2. Always write the unit in the answer box (e.g. £, kg, cm^2, m^3).
3. Perform reverse operation check for Section A 4-mark questions.
4. Re-read the question text to verify: Did you answer what was asked?

✎ Personal Mental Math Anchors & Formula Memory Reminders

CERT

OFFICIAL MILESTONE SIGN-OFF

Examination Readiness Verification



Dedication and systematic practice unlock lifelong career progression.

▶ Master Functional Skills • Certificate of Exam Readiness

This certifies that

_____

has completed all 6 Official-Style Mock Examinations and demonstrated competence across all Pearson Edexcel Functional Skills Mathematics Level 2 standards.

Lead Tutor: Abdul

Date Verified: _____

Average Mock Score: _____ / 64

Exam Centre: Oxford Health OH

✎ Final Examination Tips & Formula Memory Reminders

EXAM

FINAL REVISION • EXAM HALL PROTOCOL

Examination Day Protocol & Equipment Checklist

▶ Pearson Edexcel Functional Skills Mathematics Protocol

Ensure you are fully equipped and mentally prepared before sitting your Pearson Edexcel Level 2 examination:

- **Permitted Equipment: Black ballpoint pens, pencil, eraser, ruler (30 cm), scientific calculator (Section B only), and protractor.**
- **Section A Non-Calculator (25 mins): Complete all arithmetic using formal column methods and grid multiplication. Perform reverse checks.**
- **Section B Calculator (1 hr 20 mins): Check display brackets and BIDMAS precedence. Write intermediate method steps for every multi-mark question.**
- **Units & Rounding: Check whether monetary answers require £ and 2 decimal places (£24.50, not £24.5). State units explicitly.**
- **Final 5-Minute Audit: Re-read each question stem to verify you have answered what was asked.**

✓ Pacing Rule

Keep a running track of your time: spend 1.5 minutes per mark in Section B.

! Method Marks

Never write only the final answer; if you make an arithmetic error, you lose all method marks!

✎ Personal Exam Routine & Final Target Notes

Realistic exam simulation for guaranteed first-time pass

Contains 8 complete, authentic Functional Skills Maths Level 2 mock exam papers. Formatted identically to official examination booklets, allowing candidates to practice under realistic timed conditions.

INSIDE THIS PUBLICATION

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- ✔ **Authentic Exam Booklet Layout:** Real question structures, multi-step problem spaces, and official answer lines.
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